Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Seed Layer deposition and via filling on glass substrate for 3D/2.5D semiconductor packaging (aspect ratio 15:1) using interposers based on glass.			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (√), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electronical Electronics		*Mechanical Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	*Semiconductor Devices*	*Surface Treatment*	*Thin-Film Fabrication Technologies*
Concept of Proposed Technology	*The focus is on TGV Interposer technology with aspect ratios up to 1:15, and exploring a novel approach to depositing a seed layer inside the TGV cavities. Normal PVD processes to achieve a continuous layer inside the via need a lot of material deposited when vias with a big diameter and high aspect rations should be filled. As an alternative ALD offers a layer by layer deposition inside the via, but is slow and limited to certain material. IZM is looking for a method which modifies conventional physical vapor deposition (PVD) that deposition proceeds layer by layer, in a manner reminiscent of atomic layer deposition (ALD). This hybrid strategy leverages PVD's directional flux and wide material palette while imposing ALD-like self-limiting steps to reduce overcoating and conformality issues. The resulting seed layer exhibits enhanced thickness control, improved gap filling afterwards, and a reduced propensity for pinch-off or void formation in long, narrow vias.*			

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	Crucially, this approach addresses the classic PVD limitation of poor penetration into deep vias, because the staged pulses accumulate material progressively deeper into the channel while maintaining surface cleanliness and passivation of the sidewalls. In contrast, pure ALD, while excellent for conformality, tends to be slow due to its sequential, self-limiting chemistry and often requires specialized equipment to manage large-area production. The proposed process can operate at moderate pressures and temperatures compatible with standard track or cluster tool configurations, enabling tighter process integration with subsequent interconnect or filler steps.
Rationale for Joint R&D Project	*Since Fraunhofer IZM is working on silicon and glass interposer since the last 10-15 years we have great experience regarding the state of the art of interposer processing and filling technologies. We are able to fabricate interposers on Silicon and Glass wafers with different dimensions ranging from via diameter from 5-500µm and interposer thicknesses of 50-700µm. With higher aspect ratios and bigger via diameters which are in demand for glass interposers we see the need to find a new solution for applied the copper seed layer inside the via to allow electroplating of the via Together with a Korean partner we want to develop a disruptive technology for applying seed layers inside the via and on top of the wafer and bring it on a way to commercialization.*
Key Words	1. *Interposer* 2. Via Filling 3. ALD 4. 5.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	Development of a process/Technology which is able to create a continuous seed layer inside a Via. The maximum aspect ratio which should be achieved is 1:15. Via Diameters ranging from 10-40µm. And via depth of 150-600µm. Decrease of necessary copper deposition on the top side of the wafer, hence its removal is challenging during processing and excessive thicknesses tend to lose adhesion.			
R&D Target Specifications	- Via Diameter Successful Contentious Layer deposited - Aspect Ratio 1/15 Plated - Via Diameters 10-40µm - Via Depth 150µm, 600µm - Roughness Layer: <10nm - Ratio deposited Layer Thickness Top/Bottom Via: At least 1:100 or smaller			
R&D Plan	- **1st Project Year** - **1st Project Year Development Goal** - Development Interposer Design - Wafer Design – Definition of test structure's - Adhesion, Stress, Warpage, Roughness, Electrical, Reliability - Technology Verification Copper ALD/PVD Deposition - **1st Project Year Development Contents** - Tape Out First Interposer Fabrication (Via Fabrication) to allow first via filling trials - **2nd Project Year** - **2nd Project Year Development Goal** - Development of 1:15 Seed Layer Deposition Technologies - Tape Out Second Interposer (for fabrication of a full Interposer) to allow to fabricate all the desired test structures. - Metallization Seed Layer Evaluation - Conventional Approach			

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- New metallization Approach

- **2nd Project Year Development Contents**
  - **Finalized and filled Interposer**
  - **Evaluation of metallization results on First interposer**
  - **Processing of the Interposer with the desired structures to allow characterization**

- 3rd Project Year

- **3rd Project Year Development Goal**
  - Verification of the developed technology for via filling and seed layer deposition.
- **3rd Project Year Development Contents**
  - Verification of reliability warpage, stress, roughness
  - Characterization and verification of the fabricated interposer

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IZM will spearhead via-filling processes and conduct comprehensive evaluation of the seed layer quality. Their key responsibility includes benchmarking the novel approach against traditional via-filling methodologies, analyzing performance metrics, and documenting efficiency improvements.*
Expected R&R of Korean SME Partner	*Korean Partner will focus on seed layer deposition techniques and lead development of innovative processes to achieve continuous, uniform layers inside vias. Their expertise will drive project advancement toward reliable, scalable seed layer formation.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The expected outcome of this project is a newly developed, robust process for via filling that is specifically tailored to high aspect ratio, deep vias with large diameters, addressing a key limitation of current interconnect technologies. By implementing a novel seed layer deposition approach on the substrate, we aim to significantly improve the uniformity of the conductive coating inside the via, ensuring continuous coverage from top to bottom. At the same time, the process will be optimized to reduce the thickness of the seed layer on top of the wafer, thereby minimizing material consumption, mitigating top-side overburden, and reducing the need for extensive post-deposition planarization steps. This improved control over seed layer distribution is expected to enhance via filling quality, reduce defectivity, and enable more reliable metallization of challenging structures. Ultimately, the project should deliver a validated, industry-relevant process flow that can be integrated into advanced packaging and 3D integration manufacturing lines. The outcome will not only demonstrate technical feasibility but also show clear advantages compared to traditional approaches in terms of process efficiency, material usage, and performance, providing a strong foundation for future industrial adoption and further optimization.*

6. References

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(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization